AI Report

We are highly confident this text is

AI Generated

AI Probability

100%

This number is the probability that the document is AI generated, not a percentage of AI text in the document.

Plagiarism

?

The plagiarism scan was not run for this document. Go to gptzero.me to check for plagiarism.

Strength - 4/3/2026

[redacted]

FAQ addresses efficient LLM benchmarking under a fixed query budget: for a new model and a finite question bank of size (N_q), the goal is to estimate the finite-bank accuracy (\theta = \frac{1}{N_q}\sum_j z_j) and provide a confidence interval with valid frequentist coverage (e.g. The paper proposes Factorized Active Querying (FAQ) with three components: (1) a Bayesian logistic factor model fit on partially observed historical data (H) (model × question correctness); question factors ({v_j}) are fixed, and the new model's factor (u) is initialized from history and updated online via Laplace after each query. (2) Proactive Active Inference (PAI)—a finite-population extension of active inference that allows direct choice of the next question (rather than stream-based label/skip). The PAI estimator is unbiased for (\theta) and, under regularity, asymptotically normal; the resulting CI has model-free coverage regardless of factor-model correctness. (3) A hybrid sampling policy (q_t(\cdot)) mixing oracle-style variance reduction ((\propto \sqrt{\hat{p}(1-\hat{p})})) and active-learning (reduce posterior variance of (\theta)), with time-varying exploration/exploitation and (\tau)-mixing. Experiments on 4.4K models and two suites (MMLU-Pro 12K, BBH+GPQA+... 9.5K questions) show up to ~5× effective sample size (ESS) vs uniform sampling and ~1.8–2.4× vs best baselines (post-hoc selected), with coverage near 95%. Results hold under varying historical-data missingness; coverage is stable across model release date and accuracy. The paper releases code and curated datasets.

Strength

The finite-population framing is well motivated. Treating the benchmark as a fixed set and (\theta) as the estimand (rather than a superpopulation mean) matches how benchmarks are actually used and avoids i.i.d. The emphasis on valid coverage for deployment decisions is appropriate, and the paper is clear that the factor model is only for efficiency—PAI delivers model-free guarantees.

PAI is a useful contribution. Extending active inference to proactive question selection (full bank, deterministic labeling) and to the finite-population setting, with a clean unbiased estimator and martingale CLT, makes the method both theoretically grounded and practical. The contrast with "nature's order" stream-based procedures

is explained well, and the ablations (FAQ vs traditional active inference at low vs high budget) give insight into when proactive selection helps.

The factor model plus online Laplace updates are a good fit: they use historical data and partial observation naturally, and the hybrid policy (oracle + active learning, with tempering and (\tau)-mixing) is clearly derived. The experimental design—post-hoc selection of baselines to avoid understating them, ESS as the main efficiency metric, coverage audits by model release date and accuracy—is rigorous. The release of code and 4.4K-model / 21.5K-question datasets is valuable for reproducibility.

Weakness

Missing citations and comparison to closely related work. The paper cites several prior works on efficient LLM evaluation but omits three directly relevant lines that should be cited and contrasted:

Active Evaluation Acquisition (AEA) (Li et al., arXiv:2410.05952) models dependencies across evaluation prompts via Neural Processes, selects a subset of prompts per model (static or dynamic, including an RL-based acquisition policy), and predicts evaluation scores on the remaining prompts. Like FAQ, the goal is to reduce the number of evaluations while accurately estimating benchmark performance; AEA focuses on point estimation and prediction error, not confidence intervals or coverage. FAQ's contribution (valid CIs, finite-population estimand, PAI) would be clearer if the authors cited AEA and explicitly compared: same high-level goal (efficient benchmarking via subset selection and dependency modeling), but FAQ provides frequentist coverage and targets finite-bank accuracy without assuming the predictive model is correct.

Efficient benchmarking of language models (Perlitz et al., arXiv:2308.11696) shows that model rankings on HELM can be recovered by evaluating on a fraction of examples and uses stratified random sampling to keep dataset representation. This is a natural baseline for "evaluate on a subset"; the paper should cite it and state how FAQ differs (adaptive selection, historical factor model, and valid CIs vs ranking/point estimates).

tinyBenchmarks (Polo et al., arXiv:2402.14992) reduces evaluation cost by selecting a small curated set of examples (e.g. Again, the goal overlaps with FAQ (fewer evaluations, accurate performance estimate); tinyBenchmarks does not provide confidence intervals or coverage guarantees. static vs adaptive selection, coverage vs point estimation, factor model vs IRT/clustering) would better position FAQ.

Would you add citations and a short comparison to Active Evaluation Acquisition (2410.05952), Perlitz et al. efficient benchmarking (2308.11696), and tinyBenchmarks (2402.14992), clarifying how FAQ's goals (valid CIs, finite-bank (\theta)) and method (PAI, factor model) differ from theirs?

2.5%–25% of 12K), how often does sampling with replacement select the same question more than once, and have you considered or tried a principled without-replacement variant that preserves coverage?

How do you recommend choosing the amount of historical data (and missingness level) in practice when deploying FAQ on a new benchmark?

Sentences that are likely AI-generated.

FAQs

What is GPTZero?

GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. Our model was trained on a large, diverse corpus of human-written and AI-generated text with support for English, Spanish, French, German, and other languages. To date, GPTZero has served over 17 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

When should I use GPTZero?

Our users have seen the use of AI-generated text proliferate into education, certification, hiring and recruitment, social writing platforms, disinformation, and beyond. We've created GPTZero as a tool to highlight the possible use of AI in writing text. In particular, we focus on classifying AI use in prose. Overall, our classifier is intended to be used to flag situations in which a conversation can be started (for example, between educators and students) to drive further inquiry and spread awareness of the risks of using AI in written work.

Does GPTZero only detect ChatGPT outputs?

No, GPTZero works robustly across a range of AI language models, including but not limited to ChatGPT, GPT-5, GPT-4, GPT-3, Gemini, Claude, and AI services based on those models.

What are the limitations of the classifier?

The nature of AI-generated content is changing constantly. As such, these results should not be used to punish students. We recommend educators to use our behind-the-scene Writing Reports as part of a holistic assessment of student work. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Instead, we recommend educators take approaches that give students the opportunity to demonstrate their understanding in a controlled environment and craft assignments that cannot be solved with AI. Our classifier is not trained to identify AI-generated text after it has been heavily modified after generation (although we estimate this is a minority of the uses for AI-generation at the moment). Currently, our classifier can sometimes flag other machine-generated or highly procedural text as AI-generated, and as such, should be used on more descriptive portions of text.

I'm an educator who has found AI-generated text by my students. What do I do?

Firstly, at GPTZero, we don't believe that any AI detector is perfect. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Nonetheless, we recommend that educators can do the following when they get a positive detection: Ask students to demonstrate their understanding in a controlled environment, whether that is through an in-person assessment, or through an editor that can track their edit history (for instance, using our Writing Reports through Google Docs). Check out our list of several recommendations on types of assignments that are difficult to solve with AI.

Ask the student if they can produce artifacts of their writing process, whether it is drafts, revision histories, or brainstorming notes. For example, if the editor they used to write the text has an edit history (such as Google Docs), and it was typed out with several edits over a reasonable period of time, it is likely the student work is authentic. You can use GPTZero's Writing Reports to replay the student's writing process, and view signals that indicate the authenticity of the work.

See if there is a history of AI-generated text in the student's work. We recommend looking for a long-term pattern of AI use, as opposed to a single instance, in order to determine whether the student is using AI.